

Capacity Eviction in Reconstruction-Trained Tokenizers

Draft of August 10, 2026 — text frozen pending review

Note to the reviewer. §6 reports the outcome of a pre-registered experiment that **has not been executed**. Every quantity in it is an empty slot, marked in red, bound to a named field of the verdict manifest; the three figures are drawn on placeholder data that is implausible by construction. Nothing in §6 is a measurement, and the section is written so that a paragraph or panel cropped out of context cannot be mistaken for one. Appendices D and E are complete and contain no pending values except where explicitly marked. Appendices A–C are not reproduced here — they shipped with the §3 export.

Why this is worth reviewing before the run. The purpose of pre-writing §6 is that the distance from “verdict lands” to “submittable” should be a mechanical substitution rather than a writing task, and that neither interpretation can be composed with an outcome in front of us. §6.5 therefore contains *both* interpretation paragraphs, written and dated 2026-08-10. Exactly one will be retained; the other will be deleted rather than softened.

What we would most like challenged. (i) Whether the `SURVIVES` paragraph in §6.5 concedes enough — it claims an existence proof on one slice and binds itself with three limits, and we would like to know whether a referee reads it as appropriately narrow or as hedged into meaninglessness. (ii) Whether the `NULL` paragraph is honest or self-serving: it argues a pre-registered null is a contribution to the record, and that is exactly the argument an author with a null would want to make. (iii) Whether any slot in §6 could be filled in more than one defensible way — the intent is that each binds to exactly one manifest field, so if a reviewer can see a judgement call, the binding is under-specified. (iv) Whether the placeholder figures are genuinely impossible to mistake for results with the watermark removed. (v) In Appendix E, whether the four-status scheme (measured / derived / *pin* / pending) is the right cut, and whether anything currently labelled a *pin* is being used as evidence anywhere in the paper. (vi) Anything in Appendix D that a reader would need in order to check a run against the pre-registration and cannot find there.

6 Results

The pre-registered run has not been executed. Every quantity in this section is an empty slot bound to a named field of the verdict manifest, and every figure is drawn on placeholder data that is implausible by construction. Nothing here is a measurement.

6.1 The pre-second-stage gate

The legibility gate of Section 3.7 runs on real data before any second-stage training is authorized, and it is a hard stop rather than a diagnostic: it requires at least 0.90 sign accuracy on *every one* of the six live microstructure dimensions, with no mean clause and no floor. Its outcome determines whether the rest of this section exists.

quantity	value
per-dimension sign accuracy, dims 7–12	[six values]
minimum across the six	[min]
gate outcome	[pass or hard stop]

6.2 Primary outcome

The five clauses of Section 4.6 are evaluated conjunctively: the primary outcome is `SURVIVES` only if none fails, and `NULL` otherwise. Two guards sit outside the clauses and are `HALT`-only. Figure 8 draws the surface.

clause	tests	outcome
1 paired CI	$\Delta\text{IR}(4 - 5)$ lower bound > 0	☐
2 MDE	$\Delta\text{IR}(4 - 5) \geq$ paired MDE	☐
3 placebo validity	$\text{IR}(4) - \text{IR}(2)$ lower bound > 0	☐
4 economic floor	materiality against the 0.5 floor	☐
5 deflated Sharpe	against SR_0 at $N = 60$ trials	☐
degeneracy guard	constant-direction book, any cell	[halt or clear]
power guard	across-seed range vs claimed effect	[halt or clear]
primary		[SURVIVES / NULL / INCONCLUSIVE]

Both specifications required by the amendment window are reported from the same artifacts, and their agreement or disagreement is stated here rather than in a footnote.

	v1.5 specification (primary)	retained alternative
primary outcome	☐	☐
specifications agree	[yes or no — a first-class finding either way]	

6.3 Effect sizes and secondary diagnostics

Information coefficient, rank information coefficient, mean absolute error and R^2 are reported as secondary diagnostics and do not enter the decision rule.

cell	1 BSQ/OHLCV	2 FSQ/OHLCV	3 BSQ/+micro	4 FSQ/+micro	5 placebo
net IR, seed mean	☐	☐	☐	☐	☐
across-seed range	☐	☐	☐	☐	☐
RankIC	☐	☐	☐	☐	☐
decision activity at κ^*	☐	☐	☐	☐	☐

The three marginals involving a BSQ arm are reported and not claimed, for the reason Section 7.2 gives, and the required disclosure ships with them verbatim.

6.4 Cost stress and seed dispersion

Figure 9 sweeps the round-trip cost around the pinned 0.30%; Figure 10 plots each cell's five seeds against the size of the effect being claimed. Both figures are built against the manifest schema and are watermarked until the run lands.

6.5 Interpretation

Both paragraphs below were written on 2026-08-10, before the run was executed and before any result existed. Exactly one will be retained; the other is deleted rather than softened. They are placed here, in advance, so that neither can be written to fit an outcome.

If the outcome is SURVIVES. A positive $\Delta\text{IR}(4 - 5)$ that clears the paired confidence interval, the minimum detectable effect, the economic floor and the deflation benchmark establishes that real trade-flow imbalance carries information beyond input capacity, and that a tokenizer built to preserve it delivers that information to a cost-aware trading decision. The claim ends there, and three limits bind it immediately. It is not a calibrated estimate of how much: Section 7.3 states that $\Delta\text{IR}(4 - 5)$ mixes information with an unquantified capacity handicap, so the measured difference is an upper bound on the information effect rather than an estimate of it. It is not evidence that our BSQ baseline is competitive, so the quantizer-axis marginals remain reported rather than claimed. And it is one exchange, one venue, one instrument class and one year of scored data, with no regime-transfer test — a positive result here is an existence proof on this slice, not a property of markets. The mechanism of Section 3 would remain a fixture result; a positive outcome is consistent with it and does not confirm it, because many differences between cells 2 and 4 could produce the same sign.

the pre-registered decision surface

conjunctive — SURVIVES only if none fails

		outcome
1 paired ci	one-sided 95% CI lower bound of $\Delta\text{IR}(4-5) > 0$	—
2 mde paired	$\Delta\text{IR}(4-5) \geq$ paired MDE (tabled 3.518 at $h=15$)	—
3 placebo validity	CI lower bound of $\text{IR}(4)-\text{IR}(2) > 0$ (placebo-independent)	—
4 economic floor	$\Delta\text{IR}(4-5) \geq 0.5$ annualized IR	—
5 dsr	deflated Sharpe ratio ≥ 0.95 over 60 enumerated trials	—

HALT-only guards — never flip SURVIVES $\leftrightarrow$ NULL degeneracy — power —	SURVIVES / NULL / INCONCLUSIVE / HALT_ADJUDICATE emitted —
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Figure 8: The decision surface, fixed before any result existed. Five clauses, evaluated conjunctively: the primary outcome is SURVIVES only if none fails, and NULL otherwise. Two guards sit outside the clauses and are HALT-only — they can refuse to report an outcome their own inputs cannot support, and can never flip SURVIVES to NULL or back. The rule strings shown are the ones persisted into the verdict manifest, not a restatement of them. **[PH: no values]** — every outcome cell is empty because the run has not been executed; each is filled from a single named field of that manifest.

If the outcome is NULL or INCONCLUSIVE. This is the outcome the design’s own arithmetic makes most likely, and we pre-committed it as publishable before any result existed, so it is reported without reframing. A null here means that under this pre-registered decision rule, at this budget, on this slice, the microstructure arm did not beat its placebo by a margin the design could resolve. It does *not* mean microstructure carries no information: the minimum detectable effect is 3.518 annualized IR against a prior of $|\text{RankIC}| = 0.027$, so the design would fail to detect a real effect of the size we actually expect, and Section 7.4 states that the effect we can detect is larger than the effect we predict. Nor does it mean the mechanism of Section 3 is wrong — the mechanism is established on a fixture and is not contingent on this outcome. What a null does establish is narrower and still worth publishing: that the pipeline, with the interface defect removed and the legibility gate passed, does not convert this microstructure signal into net information ratio at this scale. Given how much of this literature reports only positive results, a pre-registered null with its power analysis stated in advance is a contribution to the record rather than a failed experiment. Section 7.4 names what would change the answer: more scored data, a lower-cost venue, or an effect larger than our own prior.

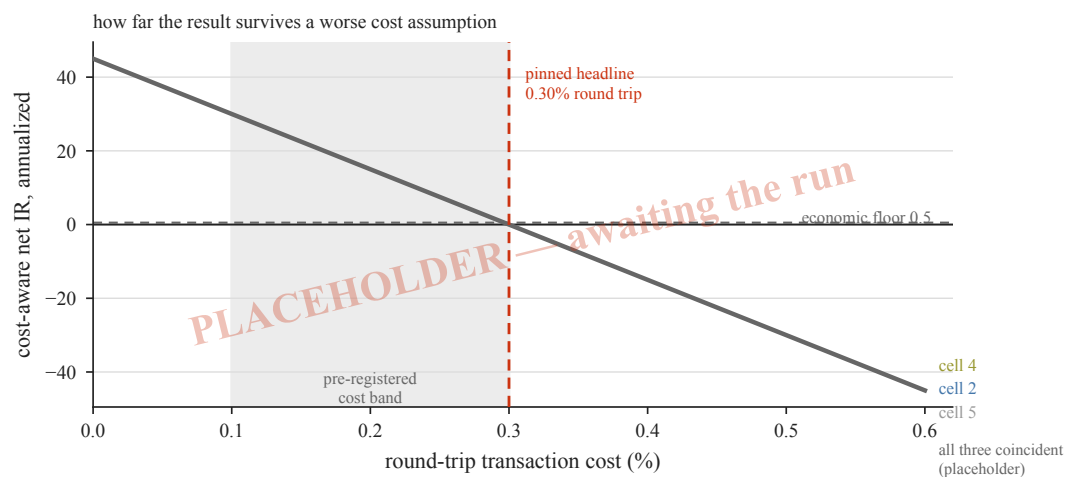


Figure 9: How far the result survives a worse cost assumption. Cost-aware net information ratio against the round-trip transaction cost, for the microstructure arm, the OHLCV-only arm and the placebo. The headline is pinned at 0.30% — the pessimistic end of the pre-registered 0.1–0.3% band — so the region to the right of that rule is stress beyond what was committed to, and the crossing of the economic floor is the quantity a reader should look for first. **[PH: one coincident curve, magnitudes $\sim \pm 45$]** — all three cells are drawn on a single line at a scale roughly two orders of magnitude above anything this design could produce, so no ordering between cells and no crossing point can be read off. The axes, the pinned rate and the floor are real; nothing else on the panel is.

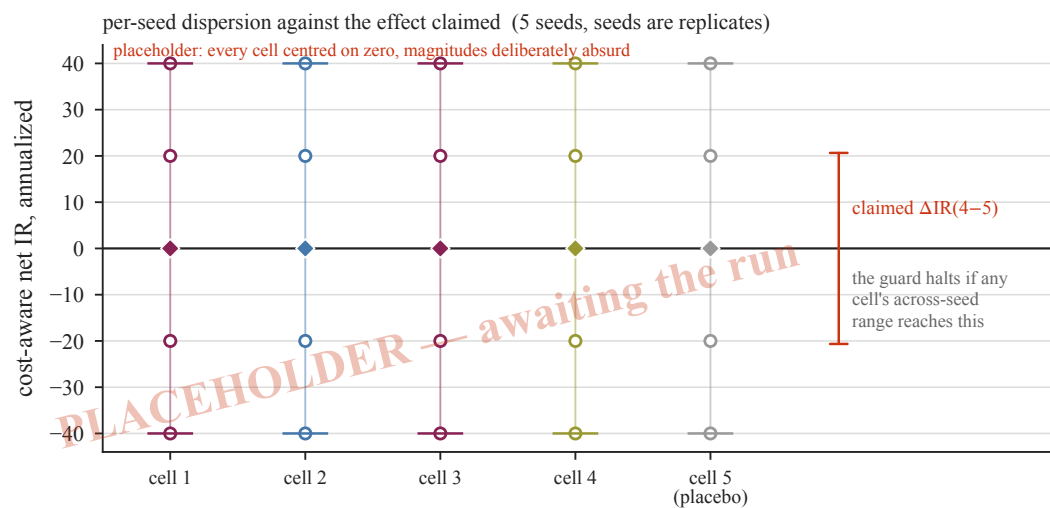


Figure 10: Seed dispersion against the effect being claimed. Each cell's five seeds are replicates of one configuration, not five configurations; open circles are seeds, the diamond is the seed mean, and the bars mark the across-seed range. The comparison that matters is the one drawn on the right: if any cell's across-seed range reaches the size of the effect being claimed, the power guard halts and the verdict is reported as adjudication-required rather than as an outcome. [PH: every cell centred on zero, magnitudes $\sim \pm 40$] — no cell is favoured and the claimed effect is drawn symmetric about zero, so the panel shows the comparison's geometry and no result.

D The pinned decision surface

Every quantity below was fixed before any of its inputs existed and is hard-asserted before any cell is scored. A run whose configuration differs from any of these raises rather than producing a down-graded result. The values are reproduced from the constants themselves, not from prose.

D.1 Experiment pins

constant	value	governs
PINNED_SYMBOLS_SHA256	60e24f59...dc32631	the evaluation universe
PINNED_SEEDS	(0, 1, 2, 3, 4)	five replicates, fixed up front
PINNED_KAPPAS	(1.0, 1.5, 2.0, 3.0)	the execution-filter grid
PINNED_HEADLINE_COST	0.0030	flat round trip, the pessimistic band end
PINNED_MONEY_SEQ_LEN	512	eval surface and training context
PINNED_BACKBONE_PARAMS	21,301,248	the realized parameter count
PINNED_STEPS_STAGE1	26,003	tokenizer budget, matched across arms
PINNED_STEPS_STAGE2	26,003	backbone budget, matched across arms
PINNED_MU_ESTIMATOR	expectation	conditional mean, not mode
PINNED_ENCODER_CAUSAL	True	forced for every cell
PINNED_FINE_POINTWISE	True	the re-specified interface
PINNED_MICRO_POINT_WEIGHT	3.0	the class weight λ
PINNED_MICRO_LEGIBILITY_MIN	0.90	the gate, on every live micro dimension
PINNED_CODEBOOK_MIN_UTILIZATION	0.95	BSQ collapse check

D.2 Decision thresholds and the deflation recipe

constant	value	governs
ECON_FLOOR_IR	0.5	clause 4 materiality floor
DSR_N_TRIALS	60	clause 5 trial count
DSR_THRESHOLD	0.95	clause 5 deflated-Sharpe threshold
trial factorization	cells $\times$ horizons $\times \kappa$	$5 \times 3 \times 4 = 60$
DSR_HORIZONS	(5, 15, 60)	the horizon axis
DSR_VAR_SR_BASIS_CELL	5	dispersion basis is the <i>placebo</i> arm
de-annualization horizon	15	the unit SR_0 is compared against
PLACEBO_DISPERSION_TRIPWIRE	1.5 $\times$	disclose if the placebo arm is over-dispersed
PINNED_BOOT	$B = 10,000$, seed 20260704, $\alpha = 0.05$	the paired bootstrap
BOOT_POWER	0.80	the MDE's power term
FRAC_NEG_BAND	[0.05, 0.95]	degeneracy guard, sign-balance leg

Two of these carry a history that Section 8.4 summarizes and the amendment log states in full. The trial count was reduced from 180 to 60 when seeds were recognized as replicates rather than configurations, and the dispersion basis was moved to the placebo arm because the earlier all-arms basis made clause 5 anti-correlated with its own hypothesis. The first tightens the criterion and the second loosens it; the net is indeterminate before the run, and Section 8.4 explains why we report it that way rather than as a single direction.

D.3 Evaluation geometry

quantity	value
window	2021-01-01 to 2025-01-01, one-minute bars
train/forward split	2023-10-20, the 0.7 point
forward blocks	6; block 0 selects κ and is excluded
headline region	blocks 1–5, $\approx$ 2024-01-01 to 2025-01-01
periods per instrument at $h = 15$	35,063
scored instruments	40 of the 200 in the lake
effective breadth	1.5
label look-forward	60 bars
serial-correlation margin	60 bars
leading embargo	120 bars

D.4 Content addressing and environment

The Gate-A anchor is a fixed evaluation of a frozen checkpoint through the full harness, re-proven twice from unpiped commands before and after any change to the evaluation instrument.

artifact	value
anchor results hash	sha256:3f86882a63dd06c78...
anchor causal sweep	420 of 420 checks
anchor dataset hash	sha256:1d3ec4f8db6686e2...
anchor tokenizer hash	sha256:ce7c8c4ae0b62c89...
anchor predictor hash	sha256:6d118ede6bfcab2e...
universe Merkle root	sha256:5dfd667d05b97bda...
lake	7,024 Parquet files, 200 symbols, 304,625,181 bars
Python	3.11.7
NumPy	2.4.6
PyTorch	2.12.1

The environment is pinned by a lockfile and load-bearing results are re-verified under it. The amendment log carries 41 distinct revision tags, each with its date, its reason, and the direction it moves the criterion, recorded before the amendment was evaluated.

E Claims audit

Every quantitative claim in this paper carries an inline comment in the $\text{\LaTeX}$ source naming the artifact it came from — 123 such comments across 44 distinct artifact paths, stripped from the submitted version and present in the released source. This appendix summarizes them by status.

Four statuses are used, and the boundaries between them are the point of the table. **Measured** means a value read from a committed receipt produced by executing something. **Derived** means computed in closed form from measured inputs, with the derivation shown at the point of use. **Pin** means a constant we chose before seeing data — a pin is not a finding, and no pin should ever be read as evidence. **Pending** means the quantity does not exist until the pre-registered run completes.

E.1 The mechanism (§3)

quantity	value	status	source
per-dim reconstruction, return channel	0.98	measured	interface re-spec receipt
per-dim reconstruction, correlated block	0.82–0.92	measured	same, dims 1–8, 10–12
per-dim reconstruction, independent dim	0.001–0.014	measured	same, dim 9, three seeds
fixture entropy match, per-dim deficit	0.4631 $\rightarrow$ 0.462	measured	filler calibration
feature-space plant, information	1.151 nats	derived	$\frac{1}{2} \ln(1 + c^2)$ at $c = 3$
feature-space plant, probe correlation	< 0.05 at all 40 evals	measured	canary v6 receipt
token-space plant, information	0.900 nats	measured	token-control receipt
token-space plant, recovery	94.4%	derived	0.8496/0.90027
per-bar legibility, original interface	0.5135	measured	id-visibility probe
chance rate	0.50	pin	probe construction
per-bar legibility, re-specified	0.9047	measured	λ -search, seed mean
legibility gate threshold	0.90	pin	conformance
bits per token, both designs	20.058	derived	$\sum \log_2 L_i$
class weight λ	3.0	pin	fitted on the fixture, then pinned

Two entries deserve a reader’s attention. The 1.151 nats is a closed-form quantity computed from a receipt constant rather than a measurement, and the derivation is shown in Section 3.3 rather than asserted. And $\lambda = 3.0$ is a pin whose value was *fitted on the synthetic fixture*; Section 7.8 states the transfer risk and names the gate that guards it.

E.2 The data (§5)

quantity	value	status	source
bars	304,625,181	measured	lake surface check
instruments	200	measured	same
symbol-months present	7,024 of 9,600	measured	lake enumeration
OHLCV-usable	303,283,972 (99.56%)	measured	ingest DQ table
microstructure-usable	300,311,536 (98.58%)	measured	same
microstructure availability	99.0%, none below 0.7	measured	same
stale / zero-volume share	0.175% / 0.376%	measured	same
delisted at ingest scan	41	measured	universe bootstrap
data ends inside the window	17	measured	lake enumeration
warm-up drop, thinnest instrument	809 of 279,930	measured	DQ spot-check
z-score half-lives	1,440 / 5,760	pin	feature config
clip range	[−5, +5]	pin	feature config
feature dimensions, total / live	16 / 13	measured	constants, mask census

E.3 The design (§4)

quantity	value	status	source
minimum detectable effect, $h = 15$	3.518 annualized IR	derived	MDE inputs receipt
economic floor	0.5	pin	verdict constants
prior, RankIC at $h = 5 / h = 15$	0.0270 / 0.0197	measured	M2 univariate screen
DSR trial count	60	pin	conformance
bits-per-bar gap, FSQ vs BSQ	0.058	derived	vocabulary arithmetic
parameter gap	69,632 (0.327%)	measured	realized counts
signed-return ACF at lag 60	0.0067 mean, 0.0191 worst	measured	embargo premise
absolute-return ACF at lag 60	0.2323 mean, 0.3604 worst	measured	same
placebo: micro–price correlation	0.2037 $\rightarrow$ 0.0005	measured	placebo mechanism doc
placebo: micro lag-1 autocorrelation	0.4165 $\rightarrow$ 0.0006	measured	same
capacity handicap in IR units	—	not quantified	Section 7.3
filter activity on real data	—	pending	no artifact records it

The last two rows are the ones a reader should carry forward. The capacity handicap is disclosed and unquantified — no cell in the pre-registration brackets it. And no eval artifact in this project records decision activity on real bars, so whether the execution filter binds there is unmeasured rather than answered elsewhere.

E.4 Verification (§8)

quantity	value	status	source
defects in the ledger	36 (19 critical)	measured	defect ledger
largest single detection layer	6 of 19 critical	derived	ledger tabulation
determinism penalty	13.175%	measured	CUDA probe report
false bit-exactness claims	49 of 49 records	measured	the D20 entry
same-seed divergence	0.5662 vs 0.99883	measured	h-sweep vs money-leg
amendment log	41 revision tags	measured	pre-registration
anchor causal sweep	420 of 420	measured	Gate-A manifest
test suite	727 passing	measured	pytest

E.5 Everything in §6

Every quantity in Section 6 is **pending**. Each is bound to a single named field of the verdict manifest, named beside the slot in the source, so that filling it is a mechanical substitution rather than a judgement. The three figures in that section are drawn on placeholder data that is implausible by construction — absent, symmetric about zero, or at magnitudes two orders above anything the design can produce — so that a cropped panel cannot be mistaken for a result. Section 6.5 carries both interpretation paragraphs, written and dated before the run.

References